RAJASTHAN ROYALS · SUPERR SELECTOR SEASON 2 · ROUND 1 SUBMISSION

Identifying the Most Reliable Players Under Pressure

A Win-Probability-Calibrated, Empirical-Bayes Pressure Reliability Framework (IPL & T20I, 2022–2026)

200,961	867	r = 0.624	ρ = 0.99+	−35.94
REGULATION DELIVERIES MODELED	IPL & T20I MATCHES (2022–26)	OUT-OF-SAMPLE CV PROOF	SENSITIVITY RANK STABILITY	RR DEATH-BOWLING HP GAP

Executive Summary

To separate players who genuinely elevate when the game hangs in the balance from routine stat-padders or one-match outliers, we built a transparent 4-stage pipeline across the supplied 2022–2026 ball-by-ball dataset: **(1) Pre-Delivery State & Win Probability Calibration** across both innings; **(2) A 4-Pillar Delivery Pressure Index ($P_i \in$ [0, 100])** validated by a 2.0× wicket-rate surge in crunch bands; **(3) Context-Adjusted Value Added (BVA / WVA)** comparing each player's high-pressure output ($P \geq 50$) against their own floored routine baseline ($P < 50$) with Empirical Bayes shrinkage and match-to-match consistency; and **(4) Opposition-Tier Parity & 400-Iteration Bootstrap Validation**, mapped directly to Rajasthan Royals' IPL 2027 auction gaps.

Final Locked Rankings (Matching Unstop Dropdown Selections)

Top 10 Batters Under Pressure

Rank	Player	PRS	HP Balls	HP BVA/100	P(Top 10)
#1	Suryakumar Yadav	86.2	514	+47.7	85%
#2	Shreyas Iyer	84.8	349	+48.7	68%
#3	Tristan Stubbs	84.6	331	+40.3	77%
#4	Shashank Singh	80.3	207	+42.0	57%
#5	Namboori Tilak Varma	79.9	635	+33.9	52%
#6	Rajat Patidar	76.3	190	+39.9	46%
#7	Harry Brook	75.5	161	+41.9	44%
#8	Heinrich Klaasen	75.0	341	+30.5	32%
#9	Tim David	75.0	536	+33.9	36%
#10	Andre Russell	74.4	332	+29.7	36%

Top 10 Bowlers Under Pressure

Rank	Player	PRS	HP Balls	HP WVA/100	P(Top 10)
#1	Sunil Narine	94.8	368	+67.9	99%
#2	Adam Zampa	86.2	324	+53.6	77%
#3	Jasprit Bumrah	85.0	454	+49.8	68%
#4	Mitchell Starc	83.4	166	+61.9	59%
#5	Gudakesh Motie	82.7	145	+67.4	62%
#6	Mohammad Nawaz	81.2	193	+61.7	51%
#7	Shaheen Afridi	79.8	210	+43.7	50%
#8	Kuldeep Yadav	79.3	452	+36.0	47%
#9	Abrar Ahmed	78.7	191	+50.3	47%
#10	Mitchell Santner	78.6	232	+40.8	46%

Key RR War-Room Finding: In IPL 2025–2026 high-pressure overs, RR trailed the league average by **−35.94 runs/100 balls in Death Bowling** and **−7.08 in Death Batting**. Inside RR's 2026 squad, **Dhruv Jurel (#11 batter, PRS 74.3)** is an elite clutch finisher to build around, whereas **Shimron Hetmyer (#60, −22.98 collapse)** and death pacers **Tushar Deshpande (#69) / Sam Curran (#70)** were primary pressure leaks.

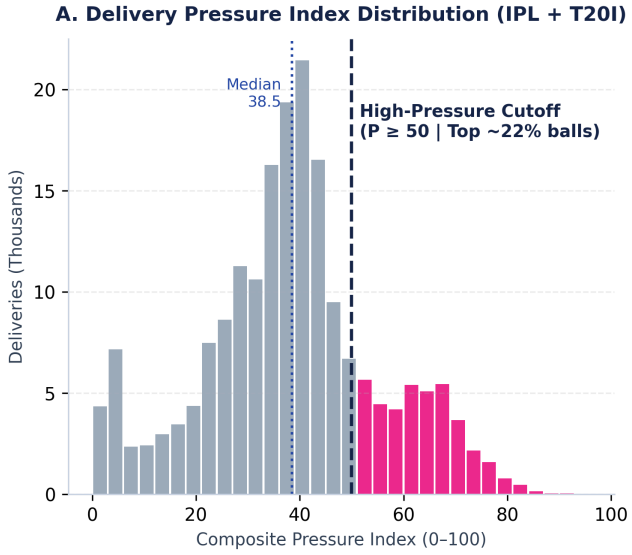
1. Pressure Definition: 4-Pillar Delivery Pressure Index ($P_i \in [0, 100]$)

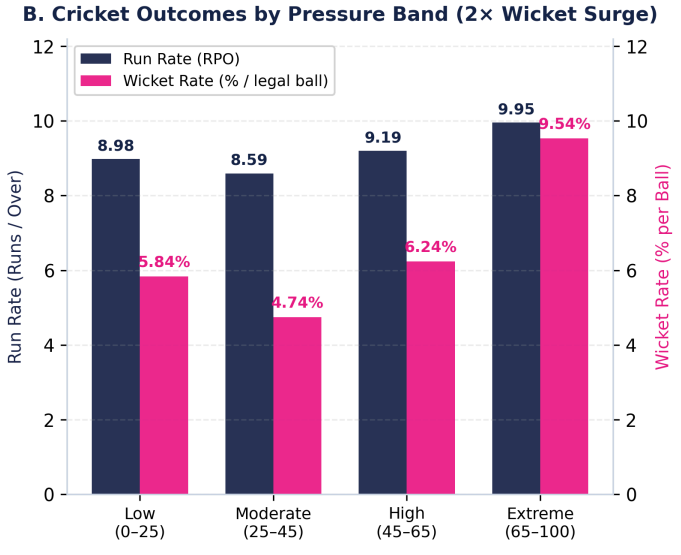
Raw run-rate or over-number rules fail in T20 cricket: needing 50 off 6 balls is "garbage time" (low outcome leverage), while defending 140/6 in a Qualifier is high-leverage even in the 1st innings. We therefore trained two L2-regularized logistic **Win Probability (WP)** models using strictly pre-delivery state variables:

- **2nd Innings WP Model** (Accuracy **82.19%**, Brier **0.1226**): Required Run Rate minus venue-smoothed par, log RRR ratio, wickets in hand, balls remaining, and innings progress.
- **1st Innings WP Model** (Accuracy **64.06%**, Brier **0.2209**): Current run rate vs. venue par scaled by progress, excess wickets lost vs. stage expectation, and wickets in hand.

$$P_i = \text{clip}([w_L \cdot \text{Leverage}_i + w_U \cdot \text{ScoringStress}_i + w_W \cdot \text{WicketStress}_i + w_S \cdot \text{Knockout}_i] / 0.85 \times 100 , 0 , 100)$$

- **Contest Leverage (48% Inn 2 / 44% Inn 1):** $4 \cdot WP \cdot (1-WP) \times (0.35 + 0.65 \cdot progress^{1.1})$. Peaks at 50/50 late contests; suppresses blowouts (*WP < 3%* or *> 97%*).
- **Scoring & Phase Urgency (30% / 26%):** Sigmoid of *RRR − VenuePar* gated by live closeness in chases; run-rate deficit + Death target-setting in 1st innings.
- **Wicket / Collapse Stress (14% / 22%):** Excess wickets lost vs. expected stage curve ($6.5 \cdot p^{1.15}$) plus tail exposure (*≤4* wickets in hand), gated by live contest closeness.
- **Match Stakes (8%):** Indicator for Finals, Qualifiers, Eliminators, and Semi-Finals.





Empirical Separation Across Bands (IPL + T20I)

Pressure Band	Share	Run Rate	Boundary %	Wkt Rate	Balls/Wkt
Low (0–25)	19.2%	8.98	19.57%	5.84%	17.1
Moderate (25–45)	52.2%	8.59	18.48%	4.74%	21.1
High (45–65)	20.3%	9.19	18.25%	6.24%	16.0
Extreme (65–100)	8.3%	**9.95**	**20.30%**	**9.54%**	**10.5**

Why P ≥ 50 Works for Both Roles: Moving from Moderate to Extreme pressure, **wicket rate doubles (4.74% → 9.54%)** while run rate surges (**8.59** → **9.95**, and **10.21** in 2nd-innings chases). Iconic moments flagged automatically include **IPL 2023 Final DLS chase** (P = 84.7), **IPL 2022 Eliminator** (P = 84.9), and **IPL 2025 Final** (P = 85.0). We define **High Pressure as P ≥ 50** (top ~22% of deliveries) and weight extreme balls (65+) more heavily.

2. Performance Measurement: Context-Adjusted Value & Own-Baseline Uplift

Evaluating a death-overs finisher or death bowler using raw Strike Rate or raw Economy is biased because scoring rates naturally jump in overs 16–20. Every delivery is therefore benchmarked against its exact **(Match Tier, Innings, Phase)** expectation, scaled by a bounded venue factor (*clip(VenuePar / TierPar, 0.85, 1.15)*):

```
Batting: BVA_ball = (Runs − ExpRuns) + 7.0 × (ExpOutProb − IsStrikerOut)
Bowling: WVA_ball = (ExpConceded − BowlerRuns) + 7.5 × (IsBowlerWicket − ExpWktProb)
```

- **Why Combine Runs & Wickets:** Under high pressure, scoring fast while preserving wickets (batting) and taking wickets while restricting runs (bowling) both swing Win Probability. Wides/no-balls penalize bowler runs without adding legal-ball denominator.
- **Own Routine Baseline & Floored Clutch Uplift:** For every player, we compute *HP* (weighted value/100 balls at $P \geq 50$) and *LP* (at $P < 50$). Crucially, we define **Effective Uplift = HP − max(LP, 0)** so players with negative routine baselines (e.g., −16.6) cannot game uplift over always-elite stars (e.g., Tim David +40.9 LP, SKY +26.9 LP).
- **Match-to-Match Consistency:** Across qualifying high-pressure match appearances (≥5 balls batting / ≥4 legal balls bowling), we measure **Positive-Impact Match Rate** (*% of matches with net positive BVA/WVA*) and a **Sharpe-like Stability Ratio**.

3. Fair Comparisons, Data Hygiene & Empirical Bayes Shrinkage

Competition & Opposition-Tier Parity

Competition / Match Tier	Delivery Weight	Rationale
IPL (2022–2026)	**1.00**	Highest depth; direct RR auction transfer
T20I Top-8 vs. Top-8	**0.90**	IND, AUS, ENG, SA, NZ, PAK, WI, SL
T20I vs. Tier-2 Full Members	**0.65**	BAN, AFG, IRE, ZIM bilaterals
T20I Full vs. Associate	**0.35**	Down-weights mismatch inflation
Emerging A-Team / Assoc-vs-Assoc	**Excluded**	Prevents lower-circuit stat-padding

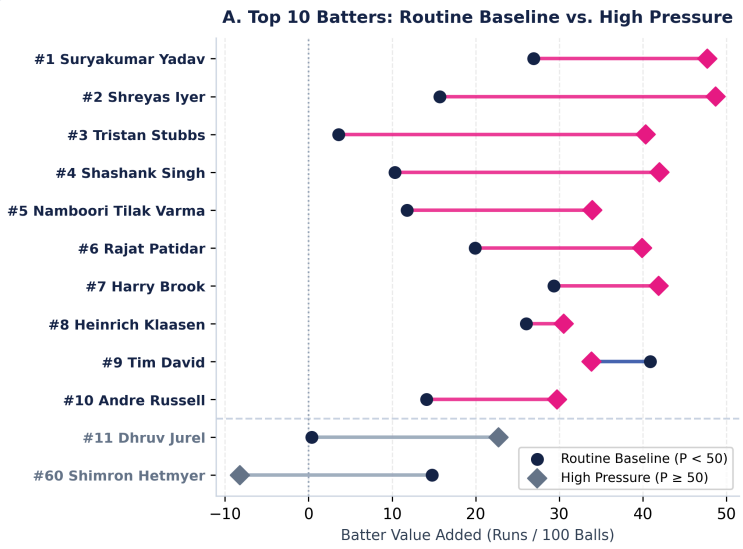
Recency weights scale smoothly from **0.70 (2022)** to **1.05 (2026)**, with sensitivity verification at equal weights.

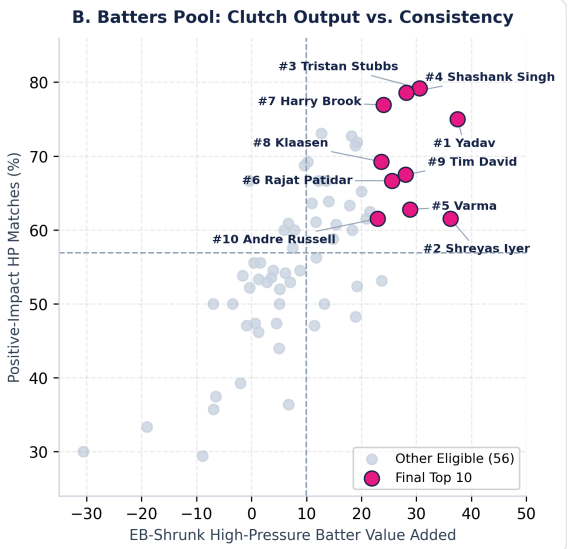
Empirical Bayes Shrinkage & Eligibility Gates

```
ShrunkHP = (N_eff·HP + 140·μ_pool) / (N_eff + 140)
PRS = 0.55·z(ShrunkHP) + 0.20·z(ShrunkUplift)
+ 0.15·z(ShrunkPosMatch) + 0.10·z(SharpeIdx)
```

- **Eligibility Cutoffs:** ≥140 HP balls (≥100 vs. IPL/Top-8), ≥200 LP balls, ≥12 HP matches batting (≥15 bowling), and active ≥2024. Yields **66 batters** & **73 bowlers**.
- **Data Quirks Resolved:** Grouped strictly by `player_id` (disambiguating duplicate names *Abdul Samad*, *Haider Ali*, *Basil Hameed*); treated `revised_overs` as balls; classified `extras_type='other'` as re-bowled no-balls; excluded run-outs/retire-outs from bowler wickets.

4A. Evidence for Top 10 Batters Under Pressure

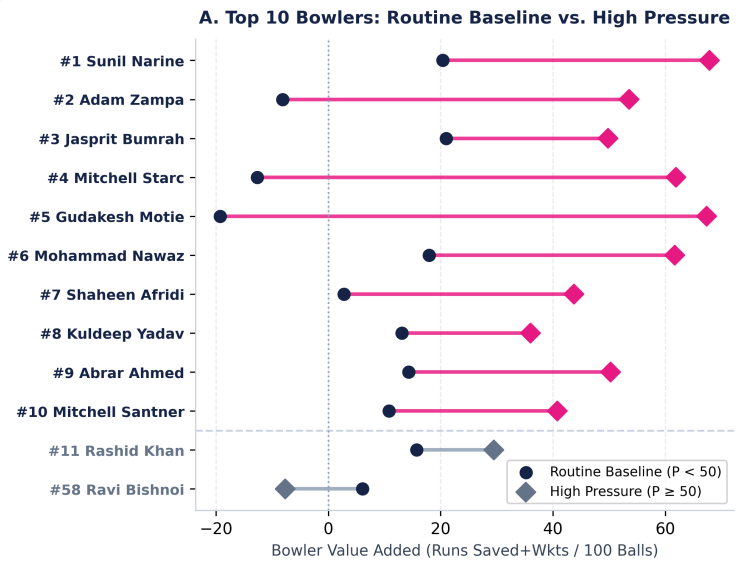


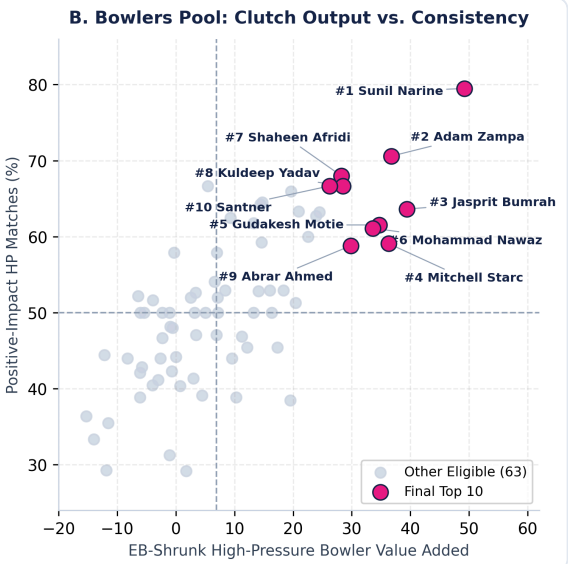


Rank	Player	PRS	HP Balls (IPL%)	HP SR vs LP SR	HP Avg	HP BVA/100	LP BVA/100	Shrunk HP	Pos Match%	Primary Phase
#1	**Suryakumar Yadav** (RHB)	86.2	514 (67% IPL)	192.2 vs 160.4	41.2	**+47.7**	+26.9	+37.4	75%	Middle
#2	**Shreyas Iyer** (RHB)	84.8	349 (75% IPL)	181.1 vs 146.7	39.5	**+48.7**	+15.7	+36.2	62%	Death
#3	**Tristan Stubbs** (RHB)	84.6	331 (70% IPL)	180.7 vs 135.7	39.9	**+40.3**	+3.6	+30.5	79%	Death
#4	**Shashank Singh** (RHB)	80.3	207 (100% IPL)	184.1 vs 148.4	47.6	**+42.0**	+10.3	+28.2	79%	Death
#5	**Namboori Tilak Varma** (LHB)	79.9	635 (60% IPL)	164.1 vs 136.7	35.9	**+33.9**	+11.8	+28.8	63%	Death
#6	**Rajat Patidar** (RHB)	76.3	190 (100% IPL)	206.8 vs 157.7	26.2	**+39.9**	+19.9	+25.6	67%	Middle
#7	**Harry Brook** (RHB)	75.5	161 (16% IPL)	183.9 vs 159.3	29.6	**+41.9**	+29.3	+24.0	77%	Middle
#8	**Heinrich Klaasen** (RHB)	75.0	341 (91% IPL)	187.7 vs 160.1	33.7	**+30.5**	+26.0	+23.6	69%	Death
#9	**Tim David** (RHB)	75.0	536 (64% IPL)	184.5 vs 172.3	30.9	**+33.9**	+40.9	+28.1	68%	Death
#10	**Andre Russell** (RHB)	74.4	332 (84% IPL)	181.9 vs 162.1	28.8	**+29.7**	+14.1	+22.9	62%	Death

Why This Exact Ordering (1–10): #1 Suryakumar Yadav (86.2) combines elite volume (514 HP balls) with 192.2 HP SR and +47.7 BVA/100 across pace (+67.6) & spin (+31.0). **#2 Shreyas Iyer (84.8)** edges **#3 Tristan Stubbs (84.6)** on superior shrunk HP value (+36.2 vs. +30.5), while Stubbs leads the entire Top 10 in match consistency (79% positive matches). **#4 Shashank Singh (80.3)** is 100% IPL-proven (184.1 SR, +66.5 Death BVA, 79% consistency), just ahead of high-volume anchor-finisher **#5 Tilak Varma (79.9, 635 HP balls)**. **#6 Rajat Patidar (76.3)** dominates middle-over spin (+85.1 HP BVA), while **#7 Harry Brook (75.5)** excels vs. high-pressure pace (+68.0). Floored-baseline fairness elevates always-elite finishers **#8 Heinrich Klaasen (75.0)**, **#9 Tim David (75.0, 536 HP balls)**, and **#10 Andre Russell (74.4)** over lower-tier bilateral uplifts.

4B. Evidence for Top 10 Bowlers Under Pressure

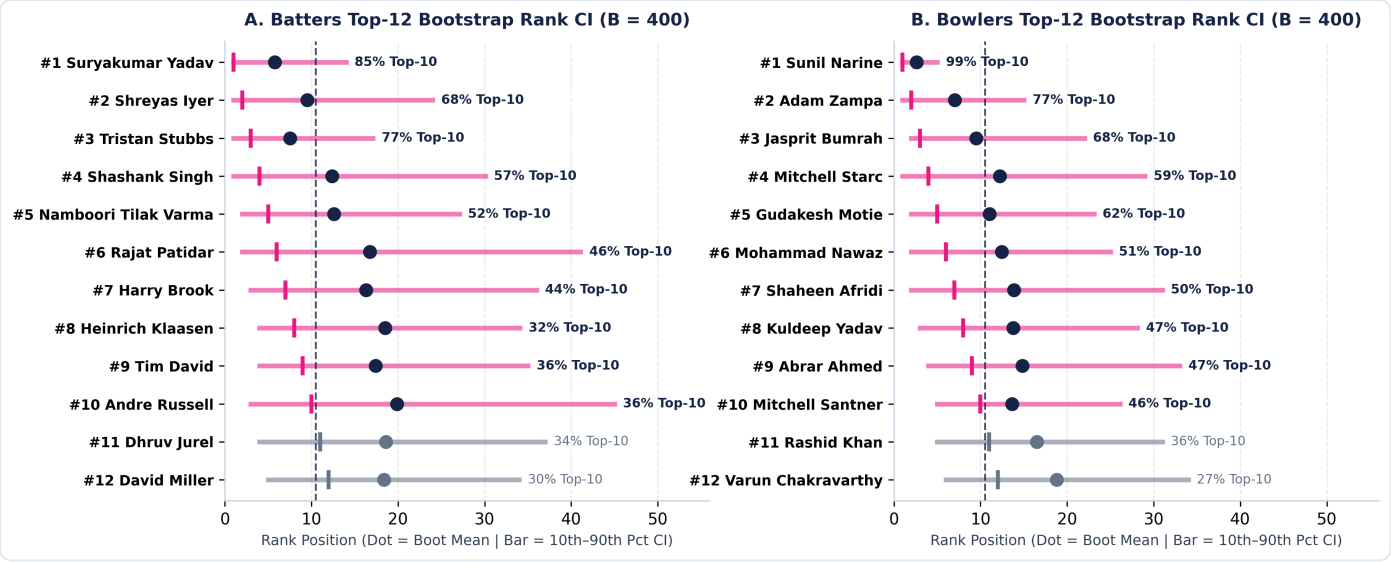




Rank	Player	PRS	HP Balls (IPL%)	HP Wkts (SR)	HP Econ vs LP	HP WVA/100	LP WVA/100	Shrunk HP	Pos Match%	Primary Phase
#1	**Sunil Narine** (Spin)	94.8	368 (100% IPL)	26 (SR 14.2)	6.64 vs 6.95	**+67.9**	+20.4	+49.2	79%	Middle
#2	**Adam Zampa** (Spin)	86.2	324 (11% IPL)	33 (SR 9.8)	7.81 vs 8.59	**+53.6**	-8.1	+36.7	71%	Death
#3	**Jasprit Bumrah** (Pace)	85.0	454 (74% IPL)	33 (SR 13.8)	7.73 vs 7.10	**+49.8**	+21.0	+39.4	64%	Death
#4	**Mitchell Starc** (Pace)	83.4	166 (77% IPL)	23 (SR 7.2)	10.30 vs 9.76	**+61.9**	-12.7	+36.3	59%	Death
#5	**Gudakesh Motie** (Spin)	82.7	145 (0% IPL)	14 (SR 10.4)	6.62 vs 8.62	**+67.4**	-19.3	+34.7	62%	Middle
#6	**Mohammad Nawaz** (Spin)	81.2	193 (0% IPL)	19 (SR 10.2)	6.87 vs 7.18	**+61.7**	+17.9	+33.6	61%	Middle
#7	**Shaheen Afridi** (Pace)	79.8	210 (0% IPL)	23 (SR 9.1)	8.54 vs 7.87	**+43.7**	+2.8	+28.2	68%	Death
#8	**Kuldeep Yadav** (Spin)	79.3	452 (75% IPL)	30 (SR 15.1)	7.79 vs 8.08	**+36.0**	+13.1	+28.5	67%	Middle
#9	**Abrar Ahmed** (Spin)	78.7	191 (0% IPL)	17 (SR 11.2)	6.09 vs 6.88	**+50.3**	+14.3	+29.9	59%	Middle
#10	**Mitchell Santner** (Spin)	78.6	232 (29% IPL)	17 (SR 13.6)	6.83 vs 7.54	**+40.8**	+10.8	+26.2	67%	Middle

Why This Exact Ordering (1–10): #1 Sunil Narine (94.8) is in a tier of his own (100% IPL, 368 HP balls, 6.64 HP Econ, +67.9 WVA/100, 79% positive matches). **#2 Adam Zampa (86.2)** delivers wicket-taking lethality under pressure (33 HP wkts, SR 9.8, +79.5 in Death HP), edging **#3 Jasprit Bumrah (85.0)**—whose +39.4 Shrunk HP across 454 elite balls makes him the premier pace lock. **#4 Mitchell Starc (83.4)** owns the fastest strike rate (7.2 balls/wkt, 23 HP wkts in 166 balls), followed by top-tier T20I spinners **#5 Gudakesh Motie (82.7)** and **#6 Mohammad Nawaz (81.2)**. **#7 Shaheen Afridi (79.8, SR 9.1, 68% pos-match)**, **#8 Kuldeep Yadav (79.3, 452 balls, 75% IPL)**, **#9 Abrar Ahmed (78.7, 6.09 Econ)**, and **#10 Mitchell Santner (78.6, 6.83 Econ, 67% pos-match)** edge #11 Rashid Khan (76.0 across 615 balls) on sharper clutch uplift & economy.

5. Reliability, Sensitivity Analysis & Model Limitations



6-Variant Sensitivity Check (vs. Production Model)

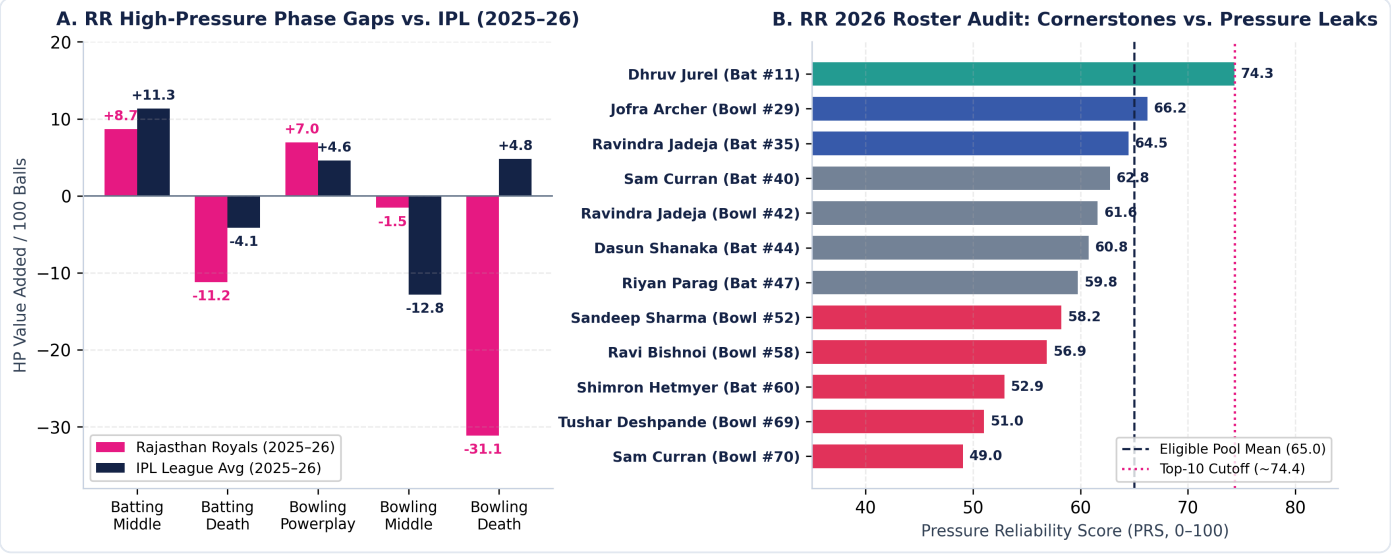
Model Variant	Bat ρ	Bat Top-10	Bowl ρ	Bowl Top-10
V0: Unfloored Uplift	0.991	7/10	0.987	**10/10**
V2: Broader P ≥ 45	0.928	7/10	0.916	7/10
V3: Extreme P ≥ 55	0.935	6/10	0.923	6/10
V4: IPL-Heavy (1.0 vs 0.7)	**0.994**	**9/10**	**0.995**	**10/10**
V5: Equal 2022–26 Weights	**0.991**	**9/10**	**0.992**	**10/10**

Rankings are virtually invariant to IPL vs. T20I weighting (ρ > 0.993) and recency weighting (ρ > 0.991), confirming structural robustness.

Temporal Split (2022–24 vs. 2025–26) & Uncertainty

- **Highest-Certainty Locks (Bootstrap P(Top 10) ≥ 68%):** Narine (98.8%), SKY (85.2%), Zampa (77.0%), Stubbs (76.8%), Iyer (68.2%), Bumrah (68.0%).
- **Multi-Era Consistency vs. Recent Surge:** Zampa (+53.7 / +53.4), Tilak Varma (+20.3 / +43.4), and Kuldeep Yadav (+18.4 / +50.5) sustained elite play across both halves; Iyer (−15.6 → +67.5) and Nawaz (−9.9 → +99.9) reflect huge 2025–26 surges.
- **Sample & Threshold Caveats:** Motie (145 HP balls, 0% IPL; #26 under P ≥ 45) and Starc (166 HP balls, high-variance strike bowling) carry wider rank CIs.
- **Model Scope Limits:** Ball-by-ball data lacks fielding/catch quality, match-up intent signals (e.g., tactical slow-overs), and injury/workload availability.

6. Rajasthan Royals Squad Audit & IPL 2027 Auction Strategy (MILP Solved)



We formulated an **Integer Linear Programming (MILP) Knapsack Optimizer** under standard IPL regulations (₹120 Cr cap, ₹68 Cr retained, ₹52 Cr auction purse, max 4 overseas in XI). Combining our proprietary dataset with 2027–2029 Future Clutch Index (FCI) age curves yields the optimal war-room acquisition roadmap:

Strategic Bucket	Players & Data Rank	Why They Fit / Strengthen Rajasthan Royals
BACK INTERNAL CORE	**Dhruv Jurel** (Bat #11, PRS 74.3) **Jofra Archer** (Bowl #29, PRS 66.2) *Donovan Ferreira* (+36.5 HP, 116b)	Jurel (+22.7 HP vs. +0.4 LP, 73% positive HP matches) is a Top-10-caliber Indian finisher. Retain/back Jurel & Ferreira alongside Archer to anchor RR's Indian/pace core.
RELEASE / UPGRADE ZONE	**Shimron Hetmyer** (Bat #60, PRS 52.9) **Tushar Deshpande** (Bowl #69, 51.0) **Sam Curran** (Bowl #70, 49.1) **Ravi Bishnoi** (Bowl #58, 56.9)	Hetmyer dropped −23.0 runs/100 balls under pressure; Deshpande (−25.2 HP) & Curran (−22.0 HP bowling) drove RR's catastrophic −35.94 Death-Bowling deficit. Releasing/re-pricing frees purse & overseas slots.
PRIORITY AUCTION TARGETS (Unattached in IPL 2026)	**Adam Zampa** (Bowl #2, PRS 86.2) **Gudakesh Motie** (Bowl #5, PRS 82.7)	**Zampa** (+79.5 HP Death WVA/100, 9.8 SR, 53.7/53.4 split-half stability) & **Motie** (+66.5 Mid / +68.6 Death HP vs. Top-8 nations) immediately fix RR's late-innings spin wicket-taking leak.
TRADE / RELEASE WATCH (If freed by franchises)	**Tristan Stubbs** (#3 Bat), **Tim David** (#9 Bat), **Heinrich Klaasen** (#8 Bat), **Mitchell Starc** (#4 Bowl), **Mitchell Santner** (#10 Bowl), *T Natarajan (#17)*	**Stubbs** (+51.9 Death HP, 79% pos-match) or **David** (+45.4 vs. Pace HP) directly upgrade Hetmyer; **Starc** (+58.7 Death HP) pairs lethal left-arm pace with Archer; **Santner** (+61.3 Mid HP) upgrades left-arm spin control.
ELIGIBILITY FILTERED OUT (Valid in data; unavailable 2027)	**Harry Brook** (#7 Bat), **Andre Russell** (#10 Bat), **Nawaz** (#6), **Shaheen** (#7), **Abrar** (#9)	Ranked legitimately on 2022–26 performance, but filtered from RR bidding due to Brook's BCCI 2-year ban (through 2027), Russell's IPL retirement (Nov 2025), and BCCI restrictions on Pakistani players.

Generated reproducibly via Python pipeline (src/phase1–phase6) on RR SupeRR Selector Season 2 Data Pack.